

Business Agent Design

Fundamentals of Agentic AI through Coding Agents

DIGB339

Tuesday 16:00–17:50

Thursday 16:00–16:50

Sukwon Hall 424

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Office hours: By appointment

Course description

An AI agent observes an environment and takes actions toward a goal. This course studies agentic AI through hands-on work with Claude Code. Codex and other coding agents will be discussed for comparison, but they are not required platforms for the course.

Students will learn how to direct agents, connect them to tools and information, evaluate their work, diagnose failures, and design systems that operate within clear limits. The course combines lectures, demonstrations, laboratories, and individual projects. Its emphasis is on building with agents while retaining human judgment and responsibility.

Projects may be practical, analytical, scientific, educational, entrepreneurial, or creative. Students choose their own domains, but must understand, test, and explain what they build.

Learning outcomes

By the end of the course, students should be able to:

- explain agents in terms of goals, environments, observations, actions, and feedback;
- use coding agents effectively within a terminal, project environment, and Git repository;
- translate a broad goal into clear, bounded, and verifiable work;
- explain how tools, skills, retrieval, MCP, and agent harnesses extend a language model;
- evaluate, debug, and improve agent-assisted work using direct evidence; and
- design and defend a substantial project while managing reliability, safety, and human oversight.

Prerequisites

Required: Successful completion of *Business Programming*, or equivalent experience with basic Python programming.

Recommended: Successful completion of *Machine Learning* and *Natural Language Processing with Deep Learning*.

The course assumes familiarity with language models. Concepts normally taught in the recommended courses may be used without review. Students should be able to write and run basic Python, interpret error messages, and inspect code written by an agent.

Required platform: Each student must maintain an individual paid Claude subscription with access to Claude Code throughout the semester. The minimum expected plan is **Claude Pro**, approximately US\$22 per month; regional pricing and taxes may vary. Codex and other agents may be discussed or demonstrated, but subscriptions to them are not required.

Course format

Most weeks include a lecture, an instructor demonstration, and a hands-on exercise. Students first work with shared examples and then apply the same ideas to projects of their own. Coding agents are used throughout the semester.

Assessment

Component	Weight
Attendance	10%
Assignments and quizzes	20%
Project 1: First agent-assisted build	20%
Project 2: Final capstone project	40%
Participation	10%
Total	100%

The attendance score begins at 10 points; each unexcused absence deducts 1 point, to a minimum of zero. The participation score begins at zero; each substantive contribution earns 1 point, up to 10 points and no more than 1 point per class meeting. Approved absences follow university policy.

Projects

All major projects are individual and open-domain.

The **first agent-assisted build** is a small project completed during the first half of the semester. It assesses the student's ability to scope a task, work iteratively with a coding agent, verify the result, and explain the finished artifact.

The **capstone project** is a more substantial artifact in a domain chosen by the student. Its 40% comprises a proposal in Week 7 (5%), a working prototype in Week 12 (5%), and the final artifact and defense in Week 16 (30%).

Projects are evaluated on purpose and scope, effective direction of agents, iterative development, verification, understanding, reliability, and the quality of the final artifact. Technical complexity and feature count are not grading objectives.

Weekly schedule

Wk.	Topic	Focus
1	Course Orientation	Course purpose, expectations, projects, and human judgment
2	Agentic AI Foundations	Goals, environments, observations, actions, feedback, and boundaries
3	Coding Environments and Git	Terminals, project environments, version control, and recoverable work
4	Task Design for Coding Agents	Clear, bounded, and verifiable assignments for agents
5	Tools and Function Calling	How agents act through files, terminals, code, and APIs
6	Skills and Reusable Workflows	Reusable instructions, references, scripts, and procedures
7	Context Engineering and RAG	Supplying relevant project and external knowledge
8	Evaluation and Verification	Tests, acceptance criteria, traces, and evidence of completion
9	Midterm Project	Demonstration and defense of the first build
10	Agentic Debugging	Reproducing failures, investigating causes, and checking repairs
11	MCP and Agent Integrations	Connecting agents to external tools, data, and services
12	Agent Harness Engineering	The runtime systems that make a language model operational
13	Agent Control Flow: Loops and Graphs	Repetition, feedback, branching, routing, and human gates
14	Safe and Responsible Agent Use	Permissions, sensitive data, untrusted inputs, and oversight
15	Optional Enrichment and Project Week	Optional clinic or enrichment; no essential new material
16	Final Project Studio and Showcase	Final testing, demonstration, and individual defense

Use of AI

Use of coding agents and other generative AI systems is expected unless an activity states otherwise. Students remain responsible for every submitted claim, artifact, action, and result.

Substantial submissions must identify the AI systems used, describe their contribution, and explain how retained outputs were verified. Students must be able to reproduce and explain their work. Fabricated project histories, test results, data, demonstrations, or disclosures are prohibited.

Data and responsible development

Do not upload confidential, proprietary, personally identifiable, or otherwise restricted information to unapproved systems. Use only data that you are permitted to process. Agents should receive only the access required for the current task, and consequential or irreversible actions must remain behind explicit human confirmation.

Students are responsible for reviewing dependencies, installation commands, external services, and licenses introduced into their projects.

Academic integrity

All Korea University academic-integrity policies apply. AI assistance does not remove the requirement that submitted work represent the student's own understanding and documented development process. Live demonstrations and spoken defenses may be used to confirm authorship and understanding.

Attendance and communication

Regular attendance is expected because laboratories, critiques, and project defenses cannot always be reproduced outside class. Official attendance requirements and approved exceptions follow university policy. Deadlines and detailed submission procedures will be announced through the course LMS.

Use the official course channel for questions about assignments and class logistics. Use email for personal or confidential matters.

Accessibility and schedule changes

Students requiring accommodations should follow university procedures and contact the appropriate office. Approved accommodations will be implemented for course activities and assessments.

Agent platforms change rapidly. The instructor may update readings, demonstrations, supported software, or laboratory details. Updates will be announced through the official course channel.